

DATA ANALYST PORTFOLIO

Ecommerce Sales Strategic Intelligence

An end-to-end analytical journey by **Aryan Mishra** — transforming 50,000+ data points into actionable growth strategies.

| The Strategic Core

The objective was to dissect global sales performance and identify hidden revenue leakages. By leveraging a high-performance tech stack, I moved data from **unstructured chaos** to **predictive clarity**.

- 🔗 Comprehensive ETL Pipeline implementation
- 📊 Advanced Segment & Regional Profiling
- ⚡ Real-time Business Intelligence Output



| The 4-Pillar Stack



Python

Pandas, Seaborn,
Matplotlib for Rigorous
EDA



SQL

Complex Joins & High-
Speed Querying



Excel

Manual Validation &
Feature Engineering



Power BI

Dynamic Visual
Storytelling Hub

Scientific EDA via Python



Pandas / Seaborn / Matplotlib

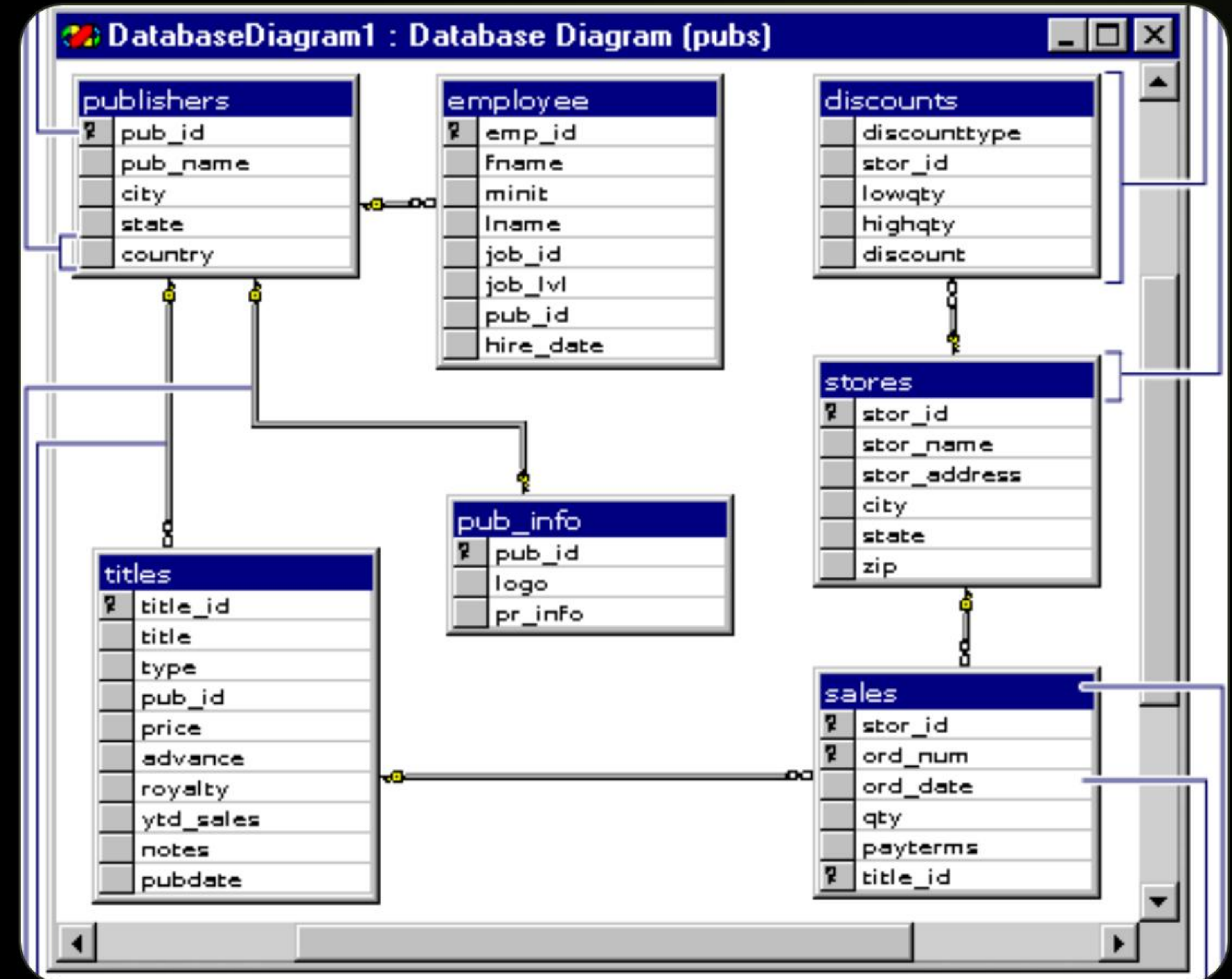
I utilized **Pandas** to engineer 50k+ rows, performing complex lambda transformations and handling outlier detection with statistical precision.

Using **Seaborn & Matplotlib**, I uncovered deep-seated correlations between order velocity and shipping modes, creating heatmaps that revealed critical logistics bottlenecks.

SQL Data Warehousing

The cleaned data was moved to a structured SQL environment to ensure ACID compliance and high-performance reporting.

```
-- Sample Query: Aggregating Performance
SELECT region, SUM(sales) AS total_revenue
FROM ecommerce_master_table
GROUP BY region
ORDER BY total_revenue DESC;
```



| The Excel Logic Layer

Before final visualization, I implemented a manual feature engineering layer in Excel to validate complex business logic and create custom segment labels.

Functionality	Impact
VLOOKUP & Power Query	Merged disparate datasets for unified customer profiling.
Manual Pivot Validation	Ensured 100% data consistency across SQL and Power BI.
Conditional Formatting	Rapid identification of negative profit anomalies.

Dynamic Visual Hub

This is the final strategic interface created in **Power BI**. It allows stakeholders to drill down into segments, categories, and dates with a single click.

🖱️ Interactive Slicers 👁️ Instant Insights

Multi Purpose PowerBI Dashboard Template



60+ Unique Visualizations

1400+ PNG Icon Download



Compatible with



| High-Level KPIs

4.74M

SUM OF SALES

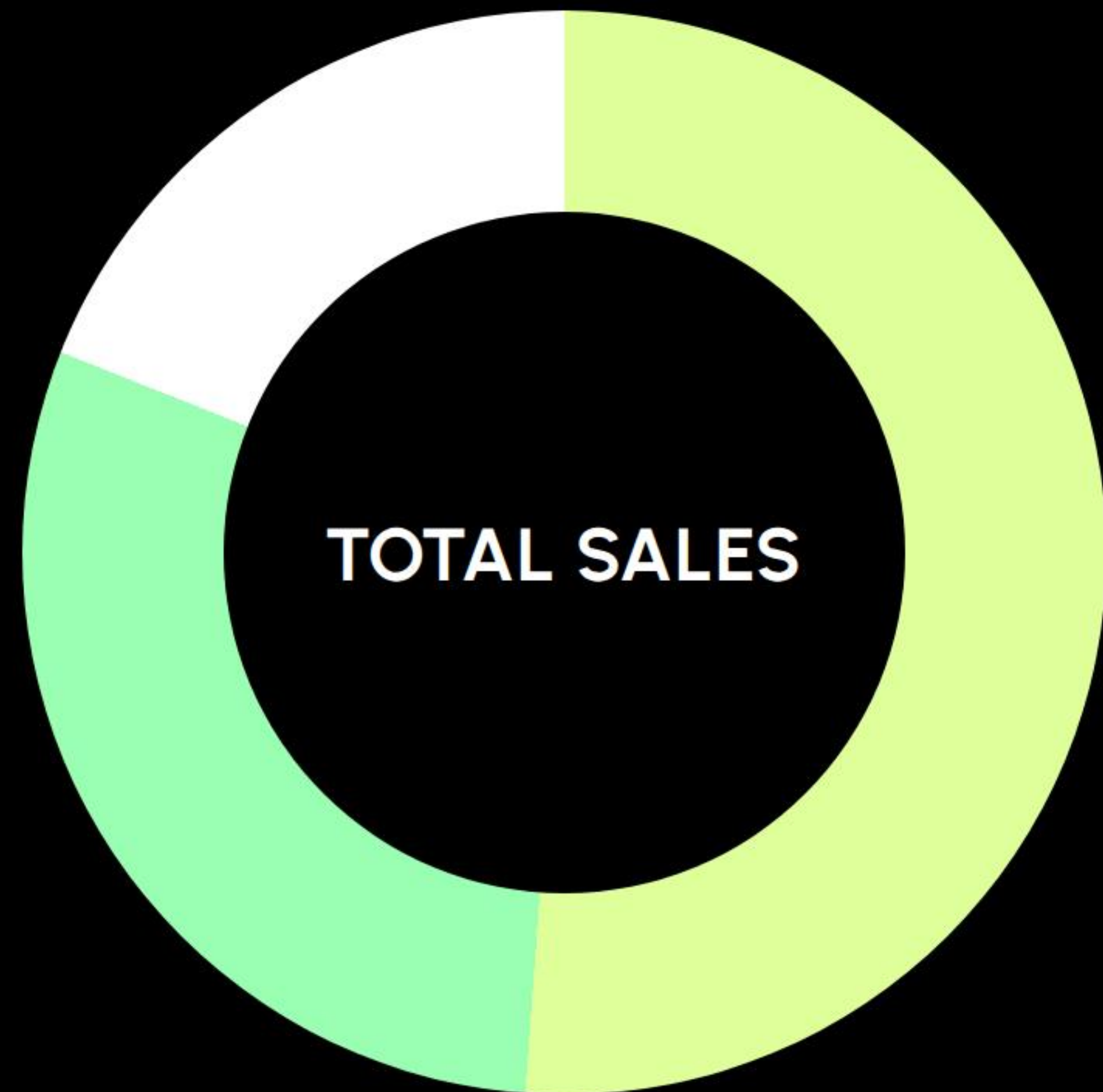
663.7K

SUM OF PROFIT

35K

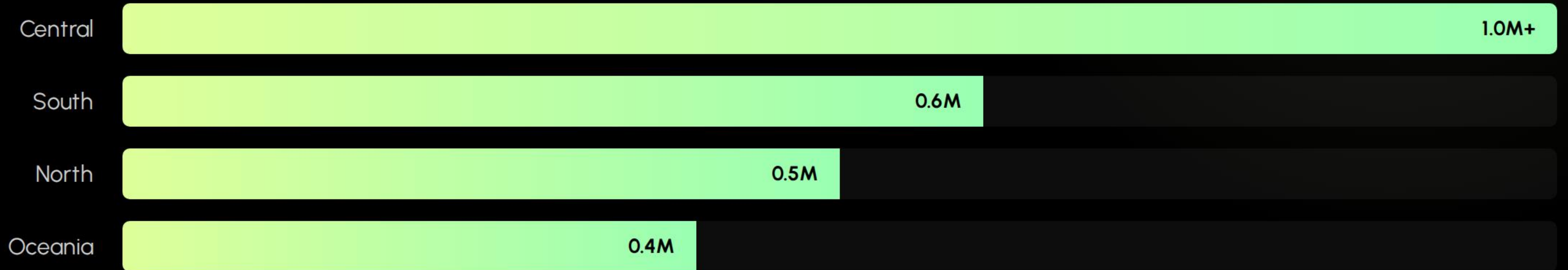
QUANTITY SOLD

| Segment Composition



-  Consumer: **51.15% (2.43M)**
-  Corporate: **29.88% (1.42M)**
-  Home Office: **18.96% (0.9M)**

| Regional Dominance



Central region exhibits the highest sales concentration, driving over 25% of global revenue.

| Final Insights

01

Market Lead

The Consumer segment is the primary engine of growth; marketing spend should be focused here.

02

Central Hub

Logistics operations in the Central region should be expanded to accommodate 1.0M+ volume.

03

Profitability

Technology category (filters applied) shows the highest average profit per order velocity.

Let's Connect

Building data-driven futures with precision.

Aryan Mishra | Data Analyst

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Image Sources



<https://static.vecteezy.com/system/resources/thumbnails/016/028/991/small/concept-of-economic-growth-chart-business-strategy-finance-and-banking-graph-and-bank-icon-on-dark-blue-background-digital-marketing-design-vector.jpg>

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